

Notes: Bloom, Eifert, Mahajan, McKenzie & Roberts (QJE, 2013)

Does Management Matter? Evidence from India

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1 Big picture

- **Question.** Can differences in management practices causally explain productivity differences across firms, especially in developing countries?
- **Setting.** A randomized field experiment in large Indian textile plants around Mumbai.
- **Intervention.** Treatment plants receive diagnostic consulting plus four months of implementation support for 38 operational practices; controls receive only diagnostic consulting.
- **Main result.** Treatment plants improve management practices and raise quality, inventory control, output, and total factor productivity.
- **Magnitude.** Productivity rises by about 17 percent in the first year, with implied annual profit gains above \$300,000.
- **Takeaway.** Management is a productive technology, and informational frictions can prevent firms from adopting profitable practices.

2 Incremental contribution of the paper

- **First large-firm randomized evidence.** The paper moves the management-productivity literature from correlations to experimental evidence in large manufacturing firms.
- **Test of best-practice management.** It shows that a common bundle of lean-style practices can be profitable, at least in this industry.
- **Mechanism-rich design.** It tracks adoption, quality defects, inventory, output, TFP, decentralization, and firm expansion.
- **Non-adoption diagnosis.** It measures whether firms failed to adopt practices because of lack of information, incorrect beliefs, owner time, or other constraints.
- **Methodological contribution.** It shows how to conduct inference in small-N, large-T firm experiments using permutation tests and large-T methods.

3 Data and measurement

3.1 Sample and intervention

The experiment covers 28 plants in 17 large family-owned textile firms around Mumbai. Treatment plants receive a diagnostic phase plus four months of implementation consulting; control plants receive only the diagnostic phase.

Interpretation: The sample focuses on complex firms where formalized management systems plausibly matter. Results are most externally valid for large, owner-managed manufacturers.

3.2 Management practice adoption

The key management variable is the share of 38 textile manufacturing practices adopted by each plant. Practices cover factory operations, quality control, inventory, loom planning, human resources, sales, and finance.

Interpretation: This is the first-stage outcome: the experiment must change practices before practices can affect productivity.

3.3 Performance outcomes

The main outcomes are quality defects, yarn inventory, output, and total factor productivity:

$$TFP_{it} = \log(Output_{it}) - 0.42 \log(Capital_{it}) - 0.58 \log(Labor_{it}).$$

Interpretation: Quality and inventory measure mechanisms; output and TFP measure performance; plant openings and decentralization capture organizational consequences.

4 Models and empirical specifications

4.1 ITT specification

$$Y_{it} = \alpha_i + \delta_t + \beta Intervention_{it} + \gamma DuringImplementation_{it} + \varepsilon_{it}.$$

Interpretation: β is the post-implementation intention-to-treat effect, comparing treatment plants to control plants with plant and week fixed effects.

4.2 Cumulative treatment specification

$$Y_{it} = \alpha_i + \delta_t + \beta CumulativeTreatment_{it} + \varepsilon_{it}.$$

Interpretation: This specification captures gradual implementation and the fact that practices accumulate over time.

5 Identification and empirical design

5.1 Random assignment and treatment contrast

Random assignment addresses the concern that better firms self-select into management adoption. Because control plants receive the diagnostic phase, the treatment contrast isolates hands-on implementation support rather than mere diagnosis.

Interpretation: The strongest causal claims are for short-run treatment effects on management adoption and plant performance within the participating firms.

5.2 Small-sample inference

The paper uses clustered bootstrap standard errors, permutation tests over the possible treatment assignments, and Ibragimov–Muller style inference.

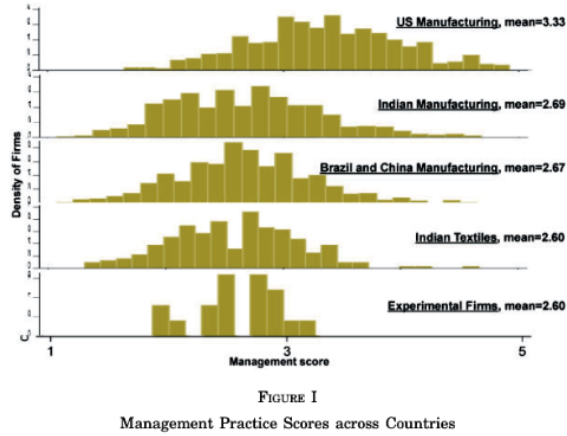
Interpretation: These methods confront the key statistical challenge: the experiment has few firms but many weekly observations.

6 Tables and figures

6.1 Figure I: Management Practice Scores across Countries; Table I: The Field Experiment Sample

Read: Figure I shows that Indian manufacturing has a thick lower tail of poor management relative to the United States. Table I shows that the experimental firms are large and complex, with treatment and control balance on observables.

Economic message: The figure motivates the experiment; the table shows that formal management systems are relevant.



(a) Figure I: Management Practice Scores across Countries.

TABLE I
THE FIELD EXPERIMENT SAMPLE

	All				Treatment Mean	Control Mean	Diff p-value
	Mean	Median	Min	Max			
Number of plants	28	n/a	n/a	n/a	19	9	n/a
Number of experimental plants	20	n/a	n/a	n/a	14	6	n/a
Number of firms	17	n/a	n/a	n/a	11	6	n/a
Plants per firm	1.65	2	1	4	1.73	1.5	0.393
Employees per firm	273	250	70	500	291	236	0.454
Employees, experimental plants	134	132	60	250	144	114	0.161
Hierarchical levels	4.4	4	3	7	4.4	4.4	0.935
Annual sales (\$m) per firm	7.45	6	1.4	15.6	7.06	8.37	0.598
Current assets (\$m) per firm	8.50	5.21	1.89	29.33	8.83	7.96	0.837
Daily mtrs, experimental plants	5,560	5,130	2,260	13,000	5,757	5,091	0.602
BVR management score	2.60	2.61	1.89	3.28	2.50	2.75	0.203
Management adoption rates	0.262	0.257	0.079	0.553	0.255	0.288	0.575
Age, experimental plant (years)	19.4	16.5	2	46	20.5	16.8	0.662
Quality defects index	5.24	3.89	0.61	16.4	4.47	7.02	0.395
Inventory (1,000 kilograms)	61.1	72.8	7.4	117.0	61.4	60.2	0.945
Output (picks, million)	23.3	25.4	6.9	32.1	22.1	25.8	0.271
Productivity (in logs)	2.90	2.90	2.12	3.59	2.91	2.86	0.869

Notes. Data provided at the plant and/or firm level depending on availability. *Number of plants* is the total number of textile plants per firm including the nonexperimental plants. *Number of experimental plants* is the total number of treatment and control plants. *Number of firms* is the number of treatment and control firms. *Plants per firm* reports the total number of textile plants per firm. Several of these firms have other businesses—for example, retail units and real estate arms—which are not included in any of the figures here. *Employees per firm* reports the number of employees across all the textile production plants, the corporate headquarters, and sales office. *Employees, experimental plants* reports the number of employees in the experimental plants. *Hierarchical levels* displays the number of reporting levels in the experimental plants—for example, a firm with workers reporting to foreman, foreman to operations manager, operations manager to the general manager, and general manager to the managing director would have five hierarchical levels. *Annual sales (\$m)* and *Current assets (\$m)* are both in 2009 US\$ million values, exchanged at 50 rupees=US\$1. *Daily mtrs, experimental plants* reports the daily meters of fabric woven in the experiment plants. Note that about 3.5 meters is required for a full suit with jacket and trousers, so the mean plant produces enough for about 1,600 suits daily. *BVR management score* is the Bloom and Van Reenen (2007) management score for the experimental plants. *Management adoption rates* are the adoption rates of the management practices listed in Appendix Table A.I in the experimental plants. *Age of experimental plant (years)* reports the age of the plant for the experimental plants. *Quality defect index* is a severity weighted measure of production quality defects. *Inventory* is the stock of yarn per intervention. *Output* is the production of fabric in picks (one pick is a single rotation of the weaving shuttle), and *Productivity* which is $\log(\text{value-added}) - 0.42 \cdot \log(\text{capital}) - 0.58 \cdot \log(\text{total hours})$. All performance measures are pooled across pre-diagnostic phase data.

(b) Table I: The Field Experiment Sample.

6.2 Figures II–IV: Dirty floors, disorganization, and excess yarn inventory

Read: The photographs show dirty factory floors, blocked movement, and yarn stored without labels, order, or damp protection.

Economic message: These figures make the mechanisms concrete: management improvements reduce repeated defects, hidden inventory, and physical disorganization.



FIGURE II
Many Parts of These Factories Were Dirty and Unsafe



FIGURE III
The Factory Floors Were Frequently Disorganized



FIGURE IV
Most Plants Had Months of Excess Yarn, Usually Spread across Multiple Locations, Often without Any Rigorous Storage System

(a) Figures II–III: Initial factory conditions.

(b) Figure IV: Excess yarn inventory.

6.3 Figure V: The Adoption of Key Textile Management Practices over Time

Read: Treatment plants increase adoption sharply; controls improve only modestly; nonexperimental plants inside treatment firms improve through within-firm spillovers.

Economic message: This is the first-stage evidence that the intervention changed management practices.

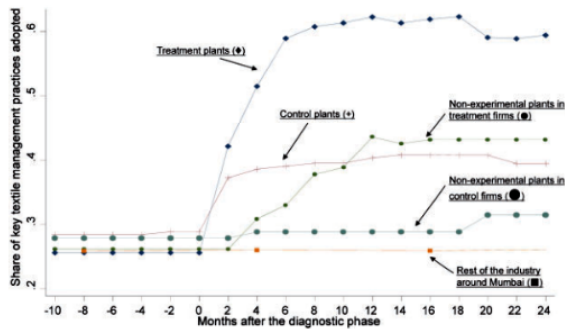


FIGURE V
The Adoption of Key Textile Management Practices over Time

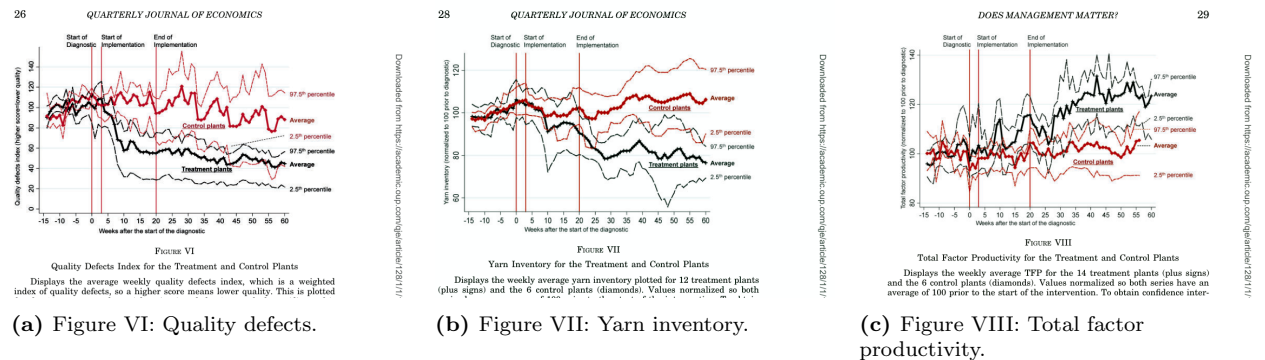
Dependent variable	(1) Quality defects	(2) Inventory	(3) Output	(4) TFP	(5) Quality defects	(6) Inventory	(7) Output	(8) TFP
Specification	ITT	ITT	ITT	ITT	Weeks of treatment	Weeks of treatment	Weeks of treatment	Weeks of treatment
Intervention _{it}	-0.564** (0.235)	-0.245** (0.117)	0.090** (0.037)	0.154* (0.084)				
During implementation _{it}	-0.260** (0.137)	-0.070 (0.093)	0.015 (0.050)	0.048 (0.050)				
Cumulative treatment _{it}					-0.022** (0.013)	-0.015** (0.005)	0.006** (0.002)	0.009** (0.004)
Small sample robustness								
Bojinovic-Muller (95% CI)	[-1.65, 0.44]	[-0.83, -0.02]	[0.05, 0.38]	[-0.014, 0.79]				
Permutation test (p-value)	.001	.000	.006	.001				
Time FEs	127	127	127	127	127	127	127	127
Plant FEs	20	18	20	18	20	18	20	18
Observations	1,807	2,052	2,393	1,831	1,807	2,052	2,393	1,831

Note: All regressions use a full set of plant and calendar week dummies. Standard errors are bootstrap clustered at the firm level. Intervention is a plant-level dummy equal to 1 after the implementation phase at treatment plants and 0 otherwise. During implementation is a dummy variable equal to 1 from the beginning of the diagnostic phase to the end of the implementation phase for all treatment plants. Cumulative treatment is the cumulative weeks of treatment since the beginning of the implementation phase in each plant (0 in both the control group and prior to the implementation phase in the treatment group). Quality defects is the log of the quality defects index (QDI), which is a weighted average score of quality defects, as higher numbers imply worse quality products (more quality defects). Inventory is the log of the tons of yarn inventory in the plant. Output is the log of the weaving production index. TFP is a plant-level total factor productivity defined as logarithm (natural) in production index less logarithm (natural) share of 0.43 less logarithm (natural) share of 0.55. ITT reports the intention to treat results from regressing the dependent variable directly on the intervention dummy. Time FEs report the number of calendar week time fixed effects. Plant FEs report the number of plant-level fixed effects. Two plants do not have any inventory on site, so an inventory data are available. Small sample robustness implements two different procedures (described in greater detail in the Appendix and Online Appendix B1) to address issues of plant heterogeneity, within-plant (and firm) correlation, and small sample concerns. Bojinovic-Muller (95% CI) report 95% confidence interval estimates from firm-by-firm parameter estimates treating the estimates as draws from independent but not identically distributed normal distributions and conduct a two-sample t-test. Permutation test reports the p-values for testing the null hypothesis that the treatment has no effect for the ITT parameter by constructing a permutation distribution of the ITT estimate using the 12,376 possible permutations of treatment assignment. These tests have exact finite sample size. *** denotes 1%, ** denotes 5%, * denotes 10% significance.

6.4 Table II: The Impact of Modern Management Practices on Plant Performance; Figures VI–VIII

Read: Table II estimates large improvements in quality, inventory, output, and TFP. The figures show the timing of quality improvement, inventory reduction, and TFP growth.

Economic message: The operational chain is coherent: better monitoring reduces defects, better inventory systems reduce excess yarn, and these changes raise productivity.



6.5 Table III: Long-Run Impact of the Experiment on Firm Size and Decentralization

Read: Better management is associated with more plants and more delegation to plant managers. Male family members explain firm size, consistent with family managerial time limiting expansion.

Economic message: The table links management to firm boundaries and reallocation, not only within-plant productivity.

LONG-RUN IMPACT OF THE EXPERIMENT ON FIRM SIZE AND DECENTRALIZATION

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TABLE IV

REASONS FOR THE NONADOPTION OF THE 18 MANAGEMENT PRACTICES (% OF ALL PRACTICES), BEFORE AND AFTER TREATMENT	Group	Management practice type		Timing relative to treatment					
Nonadoption reason				1 month before	1 month after	3 months after			
				5 months after	7 months after	9 months after			
Dependent variable	No. of plants	Experiment	z-score	Industry	Experiment	z-score			
Sample	Industry	2008–2011	2008–2011	2011	2008–2011	2008–2011			
Time period	2011	2008–2011	2008–2011	2011	2008–2011	2008–2011			
Management _i	1.040*			0.597*					
	(0.565)			(0.370)					
Male family members _i	0.210***			0.010					
	(0.065)			(0.042)					
Posttreatment _i		0.217*	0.259**		0.103**	0.171***			
		(0.122)	(0.110)		(0.049)	(0.035)			
Plant manager related _i				0.423***					
				(0.150)					
Plant manager tenure _i				0.014**					
				(0.007)					
Small sample robustness									
Permutation tests (p-value)	n/a	0.21	0.02	n/a	0.12	0.001			
Time FE _i	n/a	3	n/a	n/a	3	138			
Plant/Firm FE _i	n/a	177	121	n/a	28	108			
Observations	107	68	668	120	108	499			

Notes. The size dependent variable in columns (1)–(3) is the number of plants in the firm. The decentralization dependent variable in columns (4)–(6) is the z-score index of plant decentralization, which is the sum of the four z-scores (normalized to a mean of 0 and standard deviation of 1) individual responses over plant manager autonomy over worker hiring, spare parts purchasing authority, and does the director does not visit the factory (see Online Appendix A1 for details). Columns (1)–(3) are run at the firm level because firm size is a firm-level variable and columns (4)–(6) are run at the plant level because decentralization is a plant-level variable. Management is the adoption share of the 18 management practices started in Appendix Table A1 and discussed in Online Appendix A1, averaged across all plants within the same firm in columns (1)–(3). Male family members is the number of adult sons and brothers of the interviewed director, which includes all male family members currently working (even working in another firm) but excludes those in school or university. This is designed to measure the supply of male family members that could work in the firm. Post treatment takes the value 1 for a treatment firm after the implementation phase and 0 otherwise. Plant manager related reports if the plant manager is related to the director, including cousins, uncles, and other indirect family members. Plant manager tenure measures the number of years the plant manager has been working at the firm. Time FE report the number of calendar week time fixed effects. Firm/Firm FE reports the number of firm-level fixed effects (columns (1)–(3)) or plant-level fixed effects (columns (4)–(6)). Standard errors clustered at the firm level in all columns. Permutation test reports the p-values for testing the null hypothesis that the treatment has no effect for the ITT parameter by constructing a permutation distribution of the ITT estimate using the 12,576 possible permutations of treatment assignment. *** denotes 1%, ** denotes 5%, * denotes 10%, / denotes 15% significance.

Notes. Percentages (%) of practices not adopted by common practices are the eight practices with more than 50% initial adoption, mainly quality and downtime reduction, inventory management, and worker training. The 10 practices with less than 50% initial adoption are: worker training, inventory management, and worker training. Timing is relative to the start of the diagnostic phase. Common ITT practices in the treatment plants 18 plants, and 223 practices in the control plants 18 practices in plants. Nonadoption was ascertained every 6th week in the last shown in Online Appendix Exhibit OV, based on personal discussions with the first directors, managers, and workers.

6.6 Table IV: Reasons for the Nonadoption of the 38 Management Practices

Read: Lack of information falls sharply for treatment plants, while incorrect information and owner time constraints remain important.

Economic message: Non-adoption reflects knowledge, beliefs, and managerial bandwidth rather than only capital constraints.

6.7 Appendix Tables A.I and A.II

Read: The appendix tables list specific practices and dynamic robustness estimates.

Economic message: They show the intervention changed concrete routines rather than only a broad management index.

APPENDIX TABLE A.I The Tenma Measurement Practices Acrosses Beta					
Area	Specific practice	Preintervention level		Postintervention change	
		Treatment	Control	Treatment	Control
Factory operations	Preventive maintenance is carried out for the machine ^a	0.428	0.187	0.286	0
	Preventive maintenance is carried out per manufacturer's recommendation	0.073	0	0.073	0.187
	The shop floor is marked clearly for where each machine should be	0	0.187	0.187	0.187
	The shop floor is clear of traps and obstacles	0	0.187	0.187	0.187
	Machines downtime is recorded ^b	0.073	0.087	0.087	0
	Machines downtime reasons are recorded daily ^c	0.428	0.187	0.5	0.5
	Machines downtime analyzed at least fortnightly & action plans implemented to try to reduce that ^d	0	0.187	0.187	0
	Quality meetings take place that discuss efficiency with the production team	0	0.187	0.187	0.187
	Written procedures for setting, diving, setting, & then getting on track	0.073	0.187	0.187	0.187
	Visual aids display daily efficiency, downtime and maintenance	0.073	0.187	0.187	0.187

APPENDIX TABLE A.I (continued)					
Area	Specific practice	Preintervention level		Postintervention change	
		Treatment	Control	Treatment	Control
Quality control	Quality defects are recorded ^e	0.029	1	0.073	0
	Quality defects are recorded as a daily basis ^f	0.286	0.187	0.087	0.087
	Quality defects are recorded as a daily basis ^g	0.286	0.187	0.187	0.187
	There is an analysis and action plan based on defects data ^h	0	0.187	0.187	0.187
	There is a defect prevention system	0.073	0.187	0.187	0.187
	The production system is well defined	0.029	0.187	0.187	0.187
	Daily meetings take place that discuss defects and production	0.029	0.187	0.187	0.187
	Standard operating procedures are displayed for quality supervisors & clerical	0	0	0.187	0
	Yarn contamination (snags, loose, returns) are recorded daily ⁱ	0.029	1	0.073	0
	The clothing stock is monitored at least weekly ^j	0.029	0.187	0.187	0.187

APPENDIX TABLE A.II OLS and IV Estimation of the Effect of Management Practices on Plant Performance											
		(1)		(2)		(3)		(4)		(5)	
		OLS	IV	OLS	IV	OLS	IV	OLS	IV	OLS	IV
Dependent variable	TFP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	TFP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	TFP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	TFP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	TFP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	TFP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	TFP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	TFP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	TFP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	TFP	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

(a) Appendix Table A.I, part 1.

(b) Appendix Table A.I, part 2.

(c) Appendix Table A.II.

7 Short summary

- The paper runs a management consulting field experiment in Indian textile firms.
- Treatment plants adopt substantially more modern management practices.
- The intervention reduces quality defects, lowers inventory, raises output, and raises TFP by about 17 percent.
- Better management appears to support firm expansion and delegation.
- The main adoption barriers are lack of information, incorrect beliefs, and owner time constraints.
- The paper provides causal evidence that management practices are an important input into production.

8 Conclusion

8.1 Core conclusion

Modern management practices have large causal effects on productivity in large Indian textile firms.

8.2 Economic intuition

Management is a technology for measuring problems, assigning responsibility, and correcting errors. Better routines make the same inputs produce more usable output.

8.3 Why firms had not adopted

Firms were constrained by lack of information, skepticism about practices, and limited owner time.

8.4 Policy implications

The paper supports policies that diffuse operational know-how and improve competition and delegation institutions, but it does not imply that governments should subsidize expensive consulting for all firms.

8.5 Limitations

The sample is small, selected, and concentrated in one industry and region.

8.6 Takeaway

Managerial capital is a real production input, and productivity gaps can persist because firms fail to learn or implement profitable practices.